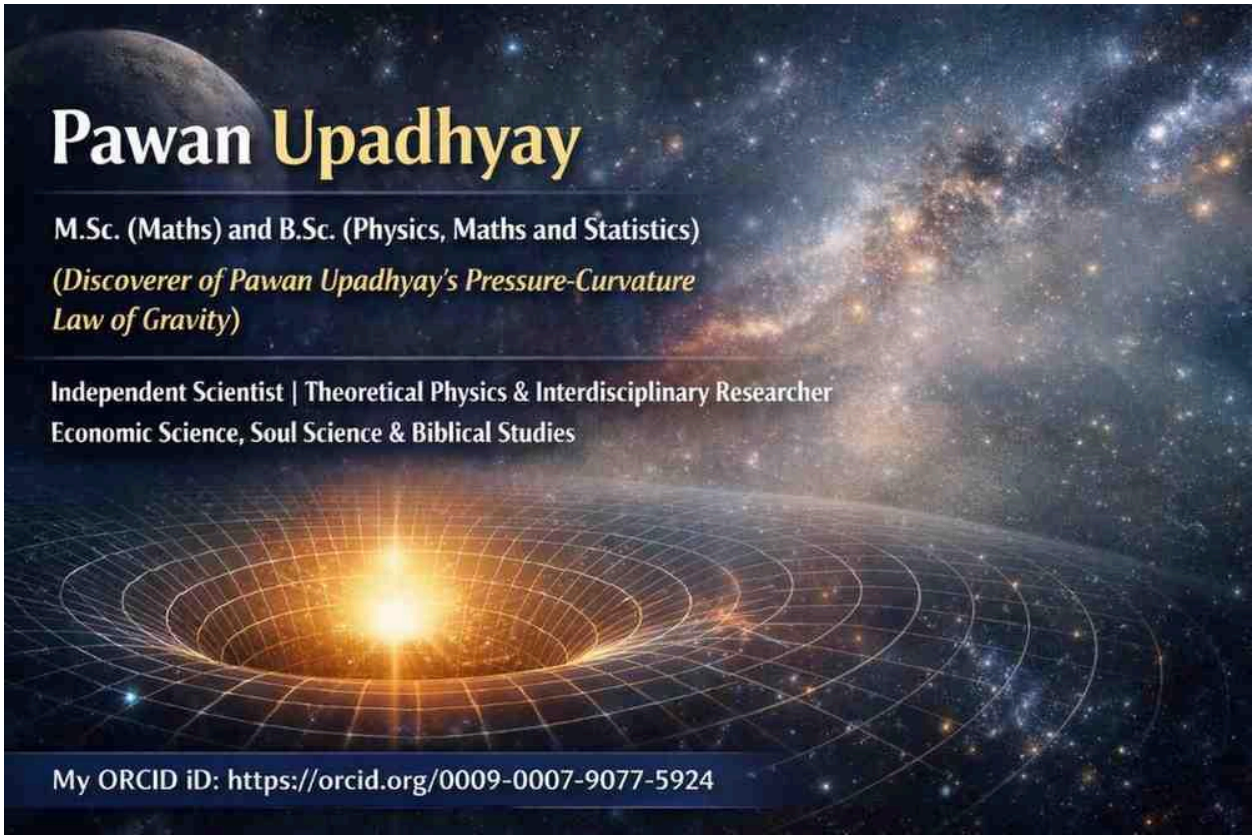
Foundations of Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)

Author: Pawan Upadhyay
Affiliation: Independent Researcher
Email: pawanupadhyay28@hotmail.com
ORCID iD: https://orcid.org/0009-0007-9077-5924
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Abstract

The nature of gravity has traditionally been described through Newtonian mechanics and later through the geometric theory of **General Relativity**.

Newton interpreted gravity as a force acting between masses, while Einstein described gravity as curvature of spacetime generated by mass–energy. Since Newton, gravity has been described as a force. Since Einstein, it has been geometry. But what causes the bending? Pawan Upadhyay's discovery answers.

This paper presents **Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)**, a theoretical interpretation in which mass–energy generates **gravitational pressure within the spacetime fabric**. Variations of this pressure create spacetime curvature, and the resulting curvature governs the motion of objects and the propagation of gravitational waves. The work outlines the **conceptual foundations, mathematical formulation, pressure–curvature relation, geodesic motion, and wave dynamics** of the PPC Law of Gravity.

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1. Introduction

Understanding gravity has been one of the central goals of physics.

In the seventeenth century, **Isaac Newton** formulated the universal law of gravitation, describing gravity as a force between masses.

In the twentieth century, **Albert Einstein** introduced **General Relativity**, which interprets gravity as curvature of spacetime produced by energy and momentum.

Although General Relativity successfully explains gravitational phenomena, the **physical mechanism that produces curvature can be interpreted in different ways**.

This Pressure-Curvature Law, discovered by Pawan Upadhyay, states that gravity is not a force of attraction nor merely a geometric effect, but the direct consequence of a pressure field generated by mass-energy acting upon the fabric of spacetime. The curvature shape is the geometric expression of this pressure force.

Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity) proposes that curvature arises due to **pressure generated by mass–energy within spacetime**.

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2. Fundamental Conceptual Points of Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)

The core principles of the PPC Law of Gravity are derived from the fundamental behavior of mass–energy and spacetime.

2.1 Mass–Energy Creates Pressure in Spacetime

Mass–energy generates pressure within the spacetime fabric.

2.2 Pressure Determines the Shape of Curvature

The distribution of gravitational pressure determines the structure of spacetime curvature.

2.3 Mass–Energy Applies Pressure on the Spacetime Fabric

Energy density associated with matter exerts pressure on spacetime.

2.4 Mass–Energy Bends Spacetime Through Pressure

The pressure produced by mass–energy deforms spacetime geometry.

2.5 Pressure Forces Determine Curvature

Forces arising from gravitational pressure gradients determine the final shape of spacetime curvature.

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3. Mathematical Framework of the PPC Law of Gravity

3.1 Mass Density

Let

ρ

represent the mass density of matter.

3.2 Energy Density

Using relativistic energy equivalence

E_d =ρ c^2

$$E_d = \rho c^2$$

where E_d represents energy density and c is the speed of light.

3.3 Gravitational Pressure

In **Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)** gravitational pressure is proportional to energy density:

P_g = ω E_d

$$P_g = \omega E_d$$

where ω is the equation of state parameter.

3.4 Pressure Gradient Force

The force produced by gravitational pressure gradients is

$$F = -\nabla P_g$$

GRAVITATIONAL PRESSURE GRADIENT EQUATION

MY PROPOSED EQUATION:

$\mathbf{F} = -\nabla P_g$

DEFINITIONS:

F = Gravitational field force density.

P_g = Gravitational pressure.

∇ = Nabla symbol, representing the spatial gradient operator.

− = Indicates that the field force density points in the direction opposite to the gravitational pressure gradient.

1. **PHYSICAL MEANING**

My proposed equation expresses a relationship between gravitational pressure and gravitational field force density.

According to my equation, gravitational field force density is determined by the spatial variation of gravitational pressure.

If gravitational pressure is uniform throughout a region, its spatial gradient is zero.

$\nabla P_g = 0$

Therefore:

$\mathbf{F} = 0$

If gravitational pressure varies from one location to another, its gradient is nonzero. My equation predicts a nonzero field force density directed toward decreasing gravitational pressure.

2. **MEANING OF THE NEGATIVE SIGN**

The negative sign in my equation indicates that gravitational field force density points opposite to the direction in which gravitational pressure increases.

If gravitational pressure decreases in a particular direction, the field force density points in that direction, according to my proposed relationship.

3. MATHEMATICAL EXAMPLE

Suppose gravitational pressure varies with position according to the following equation:

$P_g(x) = P_0 - kx$

WHERE:

P_0 = Reference gravitational pressure.

k = Positive constant.

x = Position.

Taking the derivative with respect to position:

$dP_g/dx = -k$

According to my proposed equation:

$F_x = -dP_g/dx$

Substituting the derivative:

$F_x = -(-k)$

Therefore:

$F_x = k$

In three-dimensional vector notation, if pressure varies only along the x-axis:

$\nabla P_g = -k\,\hat{\imath}$

My equation becomes:

$\mathbf{F} = -\nabla P_g$

$\mathbf{F} = -(-k\,\hat{\imath})$

$\mathbf{F} = k\,\hat{\imath}$

The resulting field force density is constant and points in the positive x-direction.

This example illustrates the mathematical behavior of my equation; it does not independently establish a physical gravitational law.

4. DIMENSIONAL ANALYSIS

Gravitational pressure P_g is measured in pascals (Pa).

$1\ \text{Pa} = 1\ \text{N/m}^2$

The spatial gradient of gravitational pressure has the following units:

$\nabla P_g = \text{Pa/m}$

Therefore:

$\nabla P_g = \text{N/m}^3$

Since F represents gravitational field force density, its SI unit is:

$F = \text{N/m}^3$

Thus, my equation is dimensionally consistent when P_g is measured in pascals and F is measured in newtons per cubic metre.

5. RELATIONSHIP TO ESTABLISHED PHYSICS

In continuum mechanics, the pressure-gradient force per unit volume is commonly expressed as:

$\mathbf{f} = -\nabla P$

My proposed equation has the same mathematical structure, with P_g representing gravitational pressure.

In standard Newtonian gravity, the gravitational force density acting on matter is expressed as:

$\mathbf{f}_g = \rho_m \mathbf{g}$

WHERE:

ρ_m = Mass density.

$\mathbf{g}$ = Gravitational acceleration.

To establish the physical validity of my proposed equation, gravitational pressure must be clearly defined and related to measurable physical quantities, such as mass density and gravitational acceleration.

A theoretical derivation and experimental testing would be required to determine whether my equation accurately describes gravitational phenomena.

6. MAIN CONCLUSIONS

1. My proposed equation establishes a mathematical relationship between gravitational pressure and gravitational field force density.

2. According to my equation, uniform gravitational pressure produces zero field force density.

3. A spatial variation in gravitational pressure produces a field force density.

4. The negative sign indicates that the field force density points opposite to the gravitational pressure gradient.

5. The equation is dimensionally consistent when gravitational pressure is measured in pascals (Pa) and field force density is measured in newtons per cubic metre (N/m^3).

6. Further theoretical development and experimental testing are required to establish whether this relationship accurately describes physical gravitational phenomena.

FINAL EQUATION:

$\mathbf{F} = -\nabla P_g$

INTERPRETATION:

My proposed equation states that gravitational field force density equals the negative spatial gradient of gravitational pressure.

NOTES ON F(x, y, z, ct) IN MY GRAVITATIONAL PRESSURE MODEL

1. **DEFINITION**

F(x, y, z, ct) represents gravitational field force density as a function of spatial coordinates and time.

Here:

F = Gravitational field force density.

x = Position coordinate along the x-axis.

y = Position coordinate along the y-axis.

z = Position coordinate along the z-axis.

c = Speed of light in vacuum.

t = Time.

ct = Product of the speed of light and time, expressed in units of length.

2. SPATIAL AND TEMPORAL DEPENDENCE

In my model, F(x, y, z, ct) represents a field quantity that may vary with spatial position and time.

The values of F may differ at different spatial locations or at different times, depending on the physical conditions described by the model.

The use of ct allows the time coordinate to have the same physical dimension as the spatial coordinates.

3. RELATIONSHIP TO MY GRAVITATIONAL PRESSURE EQUATION

My proposed equation is:

$\mathbf{F} = -\nabla P_g$

If gravitational pressure depends on spatial position and time, it may be represented as:

$P_g = P_g(x, y, z, ct)$

Therefore, the spatial-gradient equation becomes:

$\mathbf{F}(x, y, z, ct) = -\nabla P_g(x, y, z, ct)$

The nabla operator represents the spatial gradient:

$\nabla = \hat{\imath}\,\partial/\partial x + \hat{\jmath}\,\partial/\partial y + \hat{k}\,\partial/\partial z$

Consequently:

$\mathbf{F}(x, y, z, ct) = -[\hat{\imath}\,\partial P_g/\partial x + \hat{\jmath}\,\partial P_g/\partial y + \hat{k}\,\partial P_g/\partial z]$

Here, $\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ are unit vectors along the x, y and z directions, respectively.

4. IMPORTANT MATHEMATICAL DISTINCTION

The expression F(x, y, z, ct) indicates that the field depends on space and time. However, the operator ∇ in my proposed equation differentiates gravitational pressure with respect to the spatial coordinates only.

Therefore, the equation does not directly include the time derivative $\partial P_g/\partial t$.

If gravitational pressure changes with time, the field force density can change with time through its dependence on P_g, even though the spatial-gradient operator itself does not differentiate with respect to time.

5. DIMENSIONAL CONSISTENCY

The spatial coordinates x, y and z have dimensions of length.

The quantity ct also has dimensions of length because:

$[c] = L/T$

$[t] = T$

Therefore:

$[ct] = L$

The arguments x, y, z and ct can thus be expressed in a common length dimension.

However, ct is a scaled time coordinate, not an additional independent spatial direction. In relativistic physics, the conventional spacetime coordinate is often written as (x, y, z, ct).

6. FINAL FORM OF MY PROPOSED EQUATION

$\mathbf{F}(x, y, z, ct) = -\nabla P_g(x, y, z, ct)$

INTERPRETATION:

My proposed equation describes gravitational field force density as a field that may vary with spatial position and time, with its value determined by the negative spatial gradient of gravitational pressure.

This is the mathematical formulation of my proposed model. Its physical validity requires a defined gravitational-pressure field and comparison with established gravitational theory and experimental observations.

The physical force associated with gravitational pressure is given by:

$$\boxed{\mathbf{F} = -\nabla P_g}$$

Substituting the general PPC definition:

$$\mathbf{F} = -\nabla(w\rho c^2)$$

This expression:

- reproduces standard gravitational acceleration in weak fields,
- introduces **no new force law**,
- provides a **causal physical interpretation** of gravity.

For extended bodies

F_p = P_g A

$$F_p = P_g A$$

where A represents the surface area.

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4. Gravitational Pressure Field

The **gravitational pressure field** is defined as the spatial distribution of gravitational pressure generated by mass–energy in spacetime.

Mass–energy
→ energy density
→ gravitational pressure field
→ spacetime curvature.

This field represents the medium through which gravitational effects propagate.

P_g = ωE_d, where E_d = ρc² Then
P_g = ωρc². P_g is Gravitational Pressure, E_d is Energy density and ω is equation of state parameter. P_g, E_d, ω and ρ are variables.
In Special cases, ω is constant.
If values of ω are numerical.
If ω ~ 0, Then P_g ~ 0.
If ω ~ ⅓, Then P_g ~ ⅓ E_d
If ω ~ -1, Then P_g ~ -E_d
If ω ~ 1, Then P_g ~ E_d

P_g(x,y,z,ct) is gravitational pressure field.
ω(x,y,z,ct) is equation of state parameter field.
E_d(x,y,z,ct) is Energy density field.
ρ(x,y,z,ct) is mass density field.

P_g(x,y,z,ct) = ω(x,y,z,ct) E_d(x,y,z,ct)

In Strong Field Regime:
Gravitational Pressure Field is Primary. Pressure-gradient field (Four-Gravitational pressure force density field) and Four-Acceleration Field are secondary.

In Weak Field Regime:

Gravitational Pressure Field is Primary. Pressure-gradient field (Gravitational pressure force density field) and Gravitational Field are secondary.

Gravitational Field Equation (In weak field limit)

$g = -\nabla \Phi$

Where: g = Gravitational field strength, ∇ = Nabla (vector differential operator), Φ = Gravitational potential

The negative sign indicates that the gravitational field points in the direction of decreasing gravitational potential.

In Cartesian coordinates:

$g = -(\partial\Phi/\partial x\, \hat{\imath} + \partial\Phi/\partial y\, \hat{\jmath} + \partial\Phi/\partial z\, \hat{k})$

For a point mass M:

$\Phi = -GM/r$

Therefore:

$g = -(GM/r^2)\, \hat{r}$

Where: G = Universal gravitational constant M = Mass of the object r = Distance from the mass $\hat{r}$ = Outward radial unit vector

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5. Pressure–Curvature Relation

A central idea of **Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)** is that spacetime curvature is produced by gradients of gravitational pressure.

$$\mathcal{C} \propto \nabla P_g$$

where

- $\mathcal{C}$ represents curvature
- P_g represents gravitational pressure.

> **Pressure forces determine the magnitude and the shape of spacetime curvature: $E_d + P_g$ governs the magnitude of curvature, whereas ∇P_g governs its shape and the resulting orbital motion.**

This wording clearly distinguishes the two roles:

- $E_d + P_g \rightarrow$ controls the **strength (magnitude)** of curvature.
- $\nabla P_g \rightarrow$ controls the **geometry (shape)** of curvature, thereby determining the path or orbit followed by matter.

In Weak field limit (Static perfect fluid),

(E_d+3P_g) controls the strength (magnitude) of curvature.

Pressure gradient (∇P_g) controls the geometry (shape) of curvature, thereby determining the path or orbit followed by matter.

- **Magnitude of curvature:** $E_d + 3P_g$
- **Shape (geometry) of curvature:** ∇P_g

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6. Stress–Energy Tensor Representation

The combined contribution of energy density and gravitational pressure may be written as

$$T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

where

- $T_{\mu\nu}$ is the stress–energy tensor
- u_μ is four-velocity
- $g_{\mu\nu}$ is the metric tensor.

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7. Spacetime Curvature

Spacetime curvature follows

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

where $G_{\mu\nu}$ is the curvature tensor.

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Substituting PPC tensor:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left[(E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu} \right]$$

Within **Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)** curvature arises from both **energy density and gravitational pressure**.

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8. Geodesic Motion in the PPC Law of Gravity

Objects move through curved spacetime according to

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}(E_d, P_g)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0$$

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\left(g_{\mu\nu}(E_d, P_g)\right)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0.$$

indicating that geodesic motion depends on the distribution of **energy density and gravitational pressure**.

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9. Weak-Field Approximation

In weak gravitational fields the metric can be written as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

where

- $\eta_{\mu\nu}$ represents flat spacetime
- $h_{\mu\nu}$ represents small perturbations.

For a weak gravitational field, write the metric as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1.$$

In the harmonic gauge, the linearized equation is

$$\boxed{\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}}$$

where $\bar{h}_{\mu\nu}$ is the trace-reversed metric perturbation.

Weak-Field Equation

$$\nabla^2\Phi = 4\pi G\rho$$

Φ : Gravitational potential
ρ : Mass density
G : Gravitational constant
∇^2 : Laplacian operator

Static Perfect-Fluid Poisson Equation

(Weak-Field Limit)

$$\nabla^2\Phi = \frac{4\pi G}{c^2}(E_d + 3P_g)$$

Equivalently,

$$\nabla^2\Phi = 4\pi G\left(\rho + \frac{3P_g}{c^2}\right)$$

E_d : Energy density
P_g : Fluid pressure
G : Gravitational constant
c : Speed of light
∇^2 : Laplacian operator

with $\rho = \frac{E_d}{c^2}$.

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10. Gravitational Pressure Waves

Define the trace-reversed perturbation

Define the trace-reversed perturbation

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$$

Applying the harmonic gauge condition

$$\partial^{\mu}\bar{h}_{\mu\nu} = 0$$

the wave equation becomes

$$\frac{\partial^2 \bar{h}_{\mu\nu}}{\partial t^2} = c^2\nabla^2\bar{h}_{\mu\nu}$$

$$\boxed{\Box\bar{h}_{\mu\nu} = 0}$$

In **Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)** these waves correspond to **dynamic gravitational pressure disturbances propagating through spacetime**.

Note on the Scalar-Field Gravitational Wave Equation Using Capital Phi (Φ)

1. Introduction

A scalar field Φ is a mathematical field described by a single value at each point in spacetime. In gravitational theories that include a scalar degree of freedom, perturbations of this field may contribute to gravitational phenomena and, in some theories, produce a scalar gravitational-wave mode.

2. Scalar-Field Wave Equation

For a massless scalar field propagating in flat spacetime without sources, the wave equation is:

$$\Box\Phi = 0$$

Using the metric signature $(-+++)$, this becomes:

$$(1/c^2)(\partial^2\Phi/\partial t^2) - \nabla^2\Phi = 0$$

Therefore:

$$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$$

Where:

- Φ is the scalar field.
- c is the speed of light.
- ∇^2 is the spatial Laplacian.
- $\Box$ is the flat-spacetime d'Alembertian operator.

3. Scalar-Field Perturbation

Suppose the scalar field consists of a background field and a small perturbation:

$$\Phi = \Phi_0 + \delta\Phi$$

Here, Φ_0 is the background value and $\delta\Phi$ is the perturbation.

For a constant background and a massless scalar field, the perturbation satisfies:

$$\partial^2(\delta\Phi)/\partial t^2 = c^2\nabla^2(\delta\Phi)$$

This equation describes the propagation of the scalar-field disturbance at the speed of light in flat spacetime.

4. Relation to Metric Perturbation

In linearized general relativity, the metric perturbation is defined by:

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

The trace-reversed metric perturbation is:

$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h$

In vacuum and under the harmonic gauge condition, its wave equation is:

$\partial^2 \bar{h}_{\mu\nu}/\partial t^2 = c^2 \nabla^2 \bar{h}_{\mu\nu}$

The scalar equation for Φ and the tensor equation for $\bar{h}_{\mu\nu}$ have similar mathematical forms, but they describe different types of fields.

5. Physical Interpretation

A scalar-field disturbance can propagate as a wave. However, a scalar wave is not automatically a gravitational wave in the physical sense. In standard general relativity, gravitational waves have two tensor polarization states. Additional scalar gravitational-wave modes can arise in certain modified-gravity or scalar–tensor theories.

Conclusion

The equation $\Box\Phi = 0$ is the source-free wave equation for a massless scalar field in flat spacetime. Its interpretation as a scalar gravitational wave depends on the underlying gravitational theory and its field equations.

Note on the Wave Equation for Trace-Reversed Metric Perturbation

The trace-reversed metric perturbation is defined as:

$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h$

where $h_{\mu\nu}$ is the metric perturbation, $\eta_{\mu\nu}$ is the Minkowski metric, and h is the trace of the metric perturbation.

In linearized general relativity, applying the harmonic gauge condition,

$\partial_\mu \bar{h}_{\mu\nu} = 0$

reduces the vacuum Einstein field equations to the wave equation:

$\Box \bar{h}_{\mu\nu} = 0$

Equivalently,

$\partial^2 \bar{h}_{\mu\nu}/\partial t^2 = c^2 \nabla^2 \bar{h}_{\mu\nu}$

This equation describes the propagation of weak gravitational waves through vacuum at the speed of light, c.

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11. Gravitational Interaction

Gravitational interaction is defined as the motion of matter caused by spacetime curvature produced by gravitational pressure gradients.

Mass–energy
→ gravitational pressure
→ pressure gradients
→ spacetime curvature
→ motion of bodies.

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12. Dynamic Pressure Disturbances

Dynamic pressure disturbances are **time-dependent fluctuations of gravitational pressure** caused by changing mass–energy distributions.

Accelerating masses produce pressure disturbances that propagate as gravitational waves.

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13. Flow Structure of the PPC Law of Gravity

Mass Density

↓

Energy Density

↓

Gravitational Pressure

↓

Pressure Gradient

↓

Curvature via Stress-Energy Tensor

↓

Spacetime Curvature

↓

Geodesic Motion

↓

Gravitational Interaction

↓

Dynamic Pressure Disturbances

↓

Gravitational Pressure Waves

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14. Discussion

Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity) provides a physical interpretation of gravitational phenomena in which pressure generated by mass–energy produces curvature in spacetime.

This interpretation preserves the mathematical structure of relativistic gravity while emphasizing the **role of gravitational pressure as the physical mechanism behind curvature**.

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15. Conclusion

This work presents the conceptual and mathematical foundations of **Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)**.

The theory proposes that:

Mass–energy
→ generates gravitational pressure
→ pressure gradients produce curvature
→ curvature governs motion and gravitational waves.

Flow Chart of PPC Law in short

Mass Density (ρ)
↓
Energy Density ($E_d = \rho c^2$)
↓
Gravitational Pressure ($P_g = \omega E_d$)
↓
Pressure Gradient (∇P_g)

↓

Spacetime Curvature

↓

Geodesic Motion

↓

Gravitational Interaction

↓

Dynamic Pressure Disturbances

↓

Gravitational Waves (Gravitational Pressure Waves)

In General Relativity, gravity is described by the Einstein Field Equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

For a perfect fluid (model for stars and cosmology):

$$T_{\mu\nu} = \begin{pmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

This shows:

- $T^{00} = \rho c^2 = E_d$ (energy density)
- $T^{ii} = P$ (pressure components)

Therefore, **pressure contributes directly to spacetime curvature**.

Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity proposes an interpretation of gravitation in which gravitational pressure, energy density, pressure gradients, and spacetime curvature play central roles. The existing PPC framework includes the relations $E_d = \rho c^2$ and $P_g = \omega E_d$. Where P_g is Gravitational Pressure, E_d is Energy Density and ω is dimensionless equation of state parameter or dimensionless pressure scaling parameter.

1. Introduction

In **Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law)**, gravity is interpreted as a **pressure-driven phenomenon** arising from mass density and energy density. While the **pressure-gradient field force** governs motion through space, gravitational pressure also acts **directly on extended physical bodies**. This interaction gives rise to the **surface force**, which plays a crucial role in binding, deformation, tidal effects, and the physical response of matter under gravity.

Surface force does not replace the field force; rather, it complements it by describing **how gravitational pressure interacts with material bodies**.

1. Fundamental Definitions

1.1 Energy Density

In relativistic physics, mass density ρ corresponds to energy density:

$$\boxed{E_d = \rho c^2}$$

This is a **definition**, not an assumption.

1.2 Gravitational Pressure

In PPC gravity, gravitational interaction is interpreted through a **gravitational pressure field** P_g, defined generally as:

$$\boxed{P_g = w\, E_d = w\, \rho c^2}$$

where

- w is a **dimensionless pressure-scaling parameter**,
- $0 \leq w \leq 1.$

Regime	Physical system
Non-relativistic	planets, moons, stars
Relativistic	radiation-dominated
Extreme / stiff	near black holes

Physical system	Value of w
planets, moons, stars	$w \ll 1$
radiation-dominated	$w = \frac{1}{3}$
near black holes	$w = 1$

Thus,(ω) controls how much of energy density manifests as gravitational pressure.

2. Definition of Surface Force

The **surface force** in PPC gravity is defined as:

$$\boxed{F_p = P_g\, A}$$

where

- P_g is the local gravitational pressure, and
- A is the effective surface area of the body.

This force exists **only for extended bodies** and represents the direct action of gravitational pressure on matter.

3. Origin of Surface Force

Surface force is a **derived effect**, not a fundamental cause. Its origin follows the logical sequence:

$$\text{Mass density} \rightarrow \text{Pressure} \rightarrow \text{Field force} \rightarrow \text{Surface force (for extended bodies)}$$

Thus:

- **Pressure is generated by mass density**,
- **Field force establishes the pressure field**, and
- **Surface force appears when that pressure acts on a finite surface**.

Point-like test particles experience only the field force; surface force emerges only when matter has spatial extent.

3. Maximum-Pressure Limit

The relation

$$\boxed{P_g = E_d = \rho c^2}$$

It is valid **if and only if**:

$$\boxed{w = 1}$$

This corresponds to the **maximum gravitational pressure state**, applicable to:

- black holes,
- near-event-horizon regions,
- extreme relativistic collapse.

4. Ordinary Astrophysical Systems

For systems such as:

- Moon–Earth,
- Earth–Sun,
- planetary and stellar dynamics,

matter is **non-relativistic**, therefore:

$$\boxed{w \ll 1 \quad \Rightarrow \quad P_g \ll \rho c^2}$$

In these systems:

- gravitational pressure exists,
- but it is **far below the maximum-pressure limit**,
- orbital motion arises from **pressure gradients**, not from extreme pressure.

4. Distinction Between Field Force and Surface Force

- The **field force** $\mathbf{F}_{\text{field}} = -\nabla P_g$ governs **acceleration, free fall, and orbital motion**.
- The **surface force** $F_p = P_g A$ governs the **physical response of matter**.

This distinction provides conceptual clarity:

- Motion through space is governed by the field force.
- Physical interaction with gravity is governed by surface force.

5. Pressure-Gradient Force (Field Force)

The physical force associated with gravitational pressure is given by:

$$\boxed{\mathbf{F} = -\nabla P_g}$$

Substituting the general PPC definition:

$$\mathbf{F} = -\nabla (w \rho c^2)$$

This expression:

- reproduces standard gravitational acceleration in weak fields,
- introduces **no new force law**,
- provides a **causal physical interpretation** of gravity.

5. Physical Roles of Surface Force

5.1 Binding of Extended Bodies

Surface force contributes to the **structural cohesion** of planets, stars, and moons. It explains why extended bodies remain intact within strong gravitational environments and resist disruption.

5.2 Tidal Effects and Deformation

When gravitational pressure varies across a body's surface, **differential surface forces** arise. These produce:

- tides,
- stretching and compression,
- internal stress and heating.

This mechanism naturally explains tidal phenomena in systems such as Earth–Moon and star–planet interactions.

5.3 Internal Stress and Shape Formation

Surface force accounts for:

- compression within planetary interiors,
- pressure-induced spherical shapes of large bodies,
- stress distributions inside stars and compact objects.

5.4 Long-Term Orbital Evolution

While surface force does not determine orbital motion directly, it influences:

- energy dissipation,
- tidal locking,
- orbital circularization,
- long-term system stability.

6. Role in Pressure Waves

In PPC gravity, **pressure waves** arise from time-dependent variations in gravitational pressure and curvature. Surface force plays a role in how these waves:

- interact with matter,
- induce oscillations,
- generate measurable physical responses in extended bodies.

Thus, surface force mediates the **material coupling** between pressure waves and physical objects.

6. Relation to General Relativity

General Relativity already incorporates pressure through the stress–energy tensor:

$$T_{\mu\nu} = \left(\rho + \tfrac{p}{c^2}\right) u_\mu u_\nu + p\, g_{\mu\nu}.$$

In the weak-field limit:

$$\boxed{\nabla^2 \Phi = 4\pi G \left(\rho + \frac{3p}{c^2}\right) = 4\pi G \rho (1 + 3w)}$$

Thus:

- pressure is a genuine source of curvature,
- PPC gravity **does not modify GR equations**,
- it clarifies the **physical meaning** of pressure in curvature generation.

PPC EFFECTIVE STRESS–ENERGY TENSOR

$$T_{\mu\nu}^{\mathrm{PPC}} = \left(E_d + P_g\right) u_\mu u_\nu + P_g\, g_{\mu\nu}. \tag{35}$$

Where:

- E_d = effective energy density of the PPC field
- P_g = effective pressure (gravitational pressure) of the PPC field
- u_μ = 4-velocity of the fluid element $\left(u_\mu u^\mu = -1\right)$
- $g_{\mu\nu}$ = spacetime metric tensor

PPC EINSTEIN FIELD EQUATION

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}^{\mathrm{PPC}}. \tag{36}$$

Where:

- $G_{\mu\nu}$ = Einstein tensor (geometry of spacetime)
- $T_{\mu\nu}^{\mathrm{PPC}}$ = PPC effective stress–energy tensor
- G = Newton's gravitational constant
- c = Speed of light in vacuum

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Substituting PPC tensor:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left[(E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu} \right]$$

$$T_{\mu\nu}^{\text{PPC}} = (E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu} . \quad (13)$$

$$T_{\mu\nu}^{\text{PPC}} = E_d \left[(1 + \omega) u_\mu u_\nu + \omega g_{\mu\nu} \right] . \quad (14)$$

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\left(g_{\mu\nu}(E_d, P_g)\right)\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0.$$

$$E_d = \rho c^2$$

$$P_g = wE_d = w\rho c^2$$

$$\mathbf{F} = -\nabla P_g$$

$w \ll 1 \Rightarrow$ planets, moons, stars

$w = 1 \Rightarrow$ maximum-pressure (black holes)

Case	E_d	P_g	∇P_g
1. Strict vacuum	0	0	0
2. Tiny source	$\ll 1$	$\ll 1$	tiny
3. Near vacuum	$\to 0$	$\to 0$	$\to 0$

In the local rest frame,

$$T^{\mathrm{PPC}}_{\mu\nu} = \begin{pmatrix} E_d & 0 & 0 & 0 \\ 0 & P_g & 0 & 0 \\ 0 & 0 & P_g & 0 \\ 0 & 0 & 0 & P_g \end{pmatrix}$$

up to the corresponding convention-dependent factors of c.

7. What Surface Force Does Not Do

It is important to state explicitly:

- Surface force does **not** generate orbital motion
- Surface force does **not** replace spacetime curvature
- Surface force does **not** act independently of pressure fields

Orbital motion and free fall are governed by the **pressure-gradient field force** and corresponding spacetime curvature.

8. Scientific Scope

- PPC gravity is an **interpretive framework**, not a new gravitational theory.
- It preserves all predictions of Newtonian gravity and General Relativity.
- It introduces **pressure as the causal agent** behind curvature.
- Experimental and observational tests remain those of standard gravity.

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8. Placement in the PPC Framework

Fundamental causal chain

$$\text{Mass density} \rightarrow \text{Pressure} \rightarrow \text{Field force} \rightarrow \text{Curvature} \rightarrow \text{Motion} \rightarrow \text{Pressure waves}$$

Derived physical response

$$\text{Field force} \rightarrow \text{Surface force} \rightarrow \text{Binding, tides, stress, deformation}$$

9. Summary

In **Pawan Upadhyay's Pressure–Curvature Law of Gravity**, the surface force represents the **direct action of gravitational pressure on extended matter**. It governs binding, tidal effects, internal stress, and the interaction of pressure waves with physical bodies, while the pressure-gradient field force governs motion and orbital dynamics. Together, these forces provide a complete and physically intuitive description of gravitational interaction.

One-Sentence Summary

> *In PPC gravity, the surface force arises from gravitational pressure acting on extended bodies, governing binding, tides, deformation, and the physical response to pressure waves, while motion is governed by the pressure-gradient field force.*

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